

A STUDY BY PURPLE CIRCLE

THE ODYSSEY

07.17.26

ANALYZED ENTIRELY WITH MULTIMODAL TECHNIQUES



MDIG2026 · GROUP PROJECT · purplecircle

Does prior affiliation relate to movement and gaze coordination dynamics in dyadic interaction?

PRE-REGISTRATION

BalanceCorpus · Tilburg Multiscale Summer School 2026



1

Research question

Does prior affiliation relate to movement and gaze coordination dynamics in dyadic interaction ?

(And are these dynamics related to postural stability perturbation?)



Affiliation

Do Person A (cluegiver) and Person B (guesser) know each other?



Coordination

Body movement and Gaze:
change/delta scores from one frame to another for each modality



Perturbation

Wobble board vs ground condition

Literature review

Interpersonal coordination

Synchrony, coordination and mimicry →

- predicting the behavior of one's interaction partner
- reducing cognitive load
- affiliative gains between synchronized parties
- greater cooperation among participants

(Hoehl et al., 2021; Manrique et al., 2021)

Affiliation & rapport

- **Communication Accommodation Theory (CAT):** Closer relationships → More convergent vocal patterns and gestures
- Relationship characteristics precede synchrony → **synchrony reflects rapport and relationship quality**
 - Friends exhibit greater movement synchrony than strangers

(Fujiwara et al., 2020; Manrique et al., 2021)

Body movement synchronization

During unstructured interactions, this occurs at low frequencies below 0.5 Hz and can change under task demands:

- repetitive actions such as hand gestures and head nods → faster (1 Hz)
- large movements such as postural sway and leg crossing → slower (0.5 Hz)

(Fujiwara et al., 2020; Fujiwara & Daibo, 2016)

Gaze Behavior

"Orienting the gaze at the right location at the right time improves coordination"

(Khoramshahi et al., 2016)

3

Variables (the dream)



Body movement

Composite change in movement



Expressivity

Composite change in facial expressions



Gaze

Change in x, y angle



Audio

f0 change



Balance perturbation

Change in balance board gyroscope



Postural sway

Change in hip movement

Alignment

All variables must resolve to the same time series. Start/end times and alignment should already be present in the original data.

3

Variables (reality)



Body movement

Composite change in movement



Expressivity

Composite change in facial expressions



Gaze

Change in x, y angle



Audio

f0 change



Balance perturbation

Change in balance board gyroscope



Postural sway

Change in hip movement

Alignment

All variables must resolve to the same time series. Start/end times and alignment should already be present in the original data.

Operationalization

1

MdRQA

Models dyad as a single system revisiting composite states

- MdRQA on body movement time series
- Recurrence = the **dyad** returning to a previously visited joint state
- Radius set per-recording so that RR = 5%; radius varies across dyads
- Outputs: For today -> %DET, in general -> LAM, ENTR, TT

2

Windowed Multiscale Synchrony

Synchrony / coordination = SI (Synchrony Index)

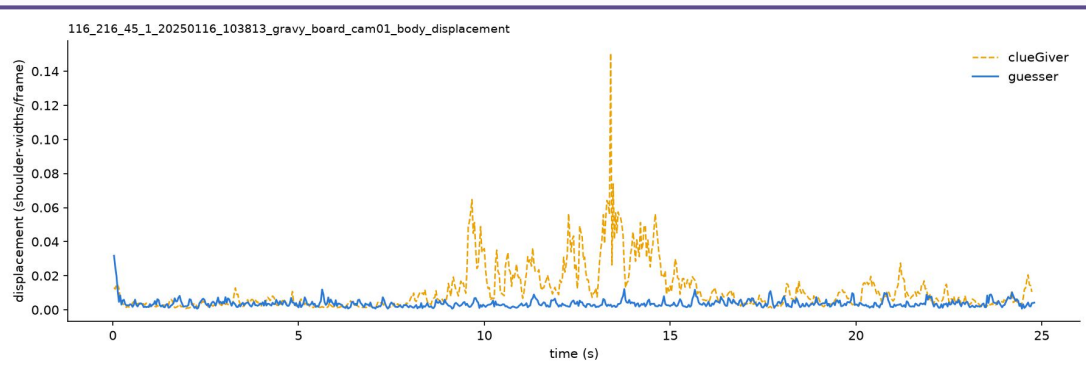
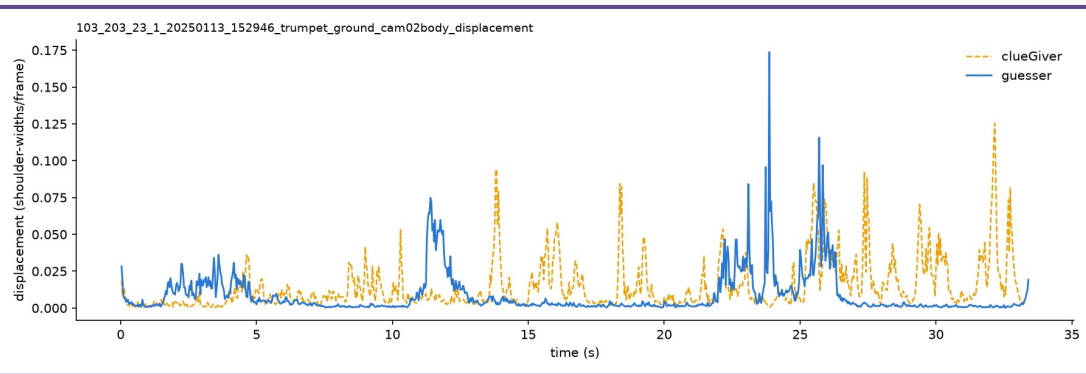
- Morlet wavelet decomposition → SI per frequency, per window
- srate = 30 Hz (OpenFace native frame rate)
- **win = 30 · minfreq = 0.25 · maxfreq = 8 Hz**
- mincycle = 5 · maxcycle = 8 · nfreqs = formula
- wavelet_length = 4

Both run cross-person: clue-giver vs guesser, within a trial.

7

Results: composite movement

proof of concept, not a hypothesis test



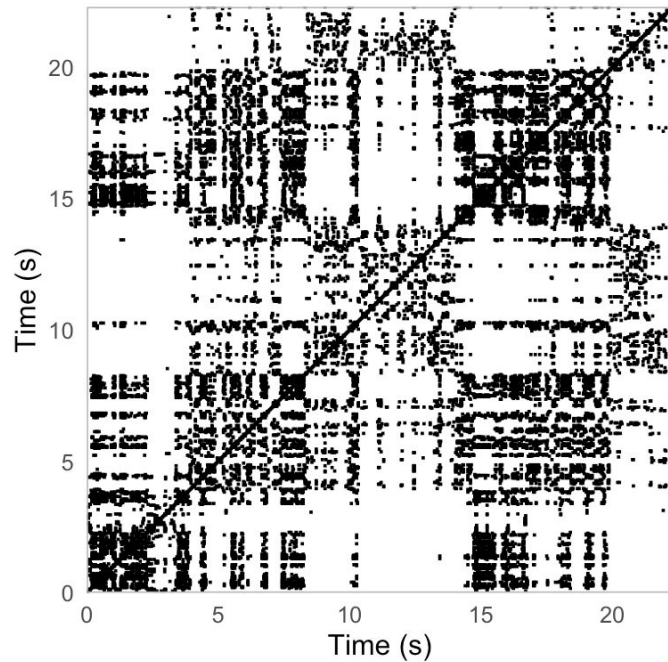
What to read here

- guesser
- clue-giver

- Non-linear
- Complex
- Limited movement in our example case :(

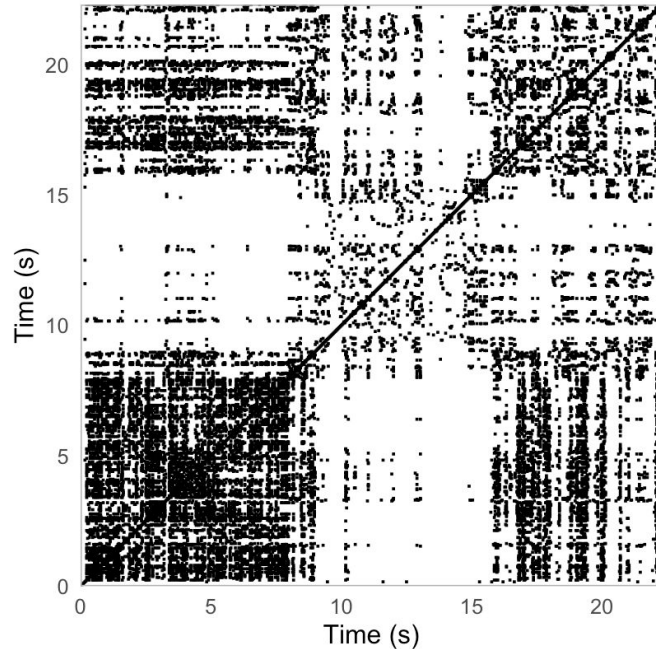
Results: MdRQA of movement composite

mdRQA recurrence plot



Unaffiliated

mdRQA recurrence plot

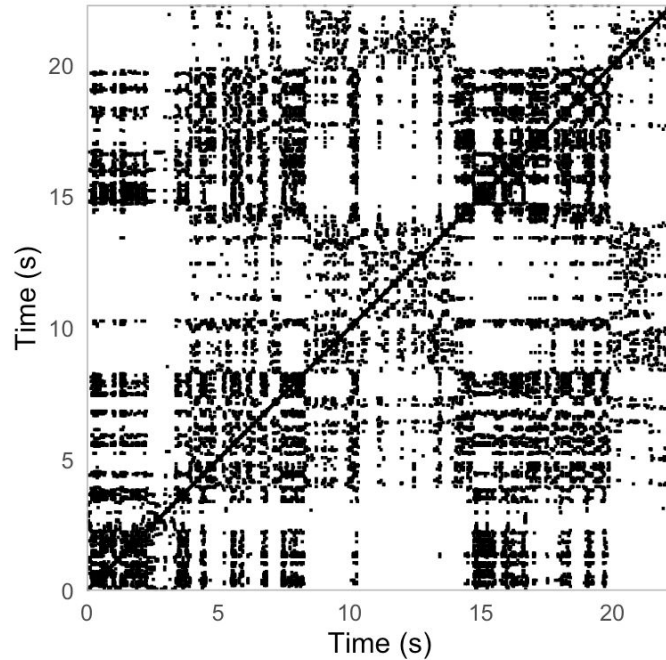


Affiliated

8 Results: MdRQA of movement composite

DET LAM
30.3037 37.37974

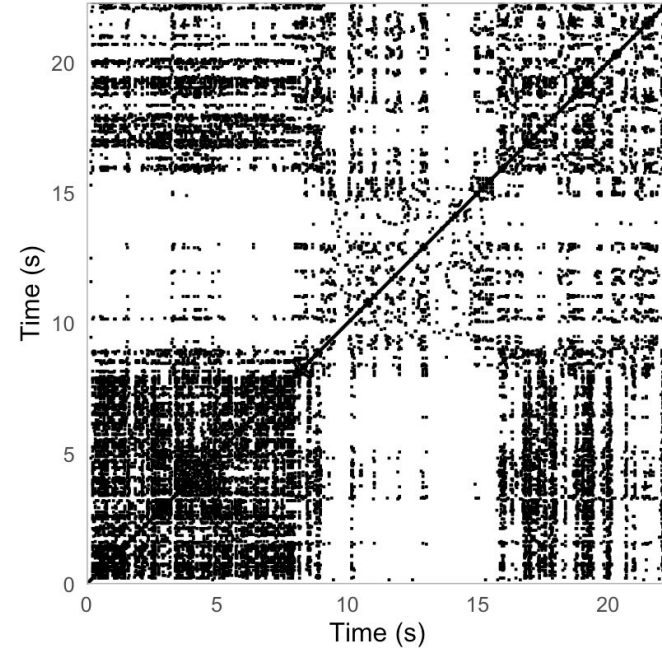
mdRQA recurrence plot



Unaffiliated

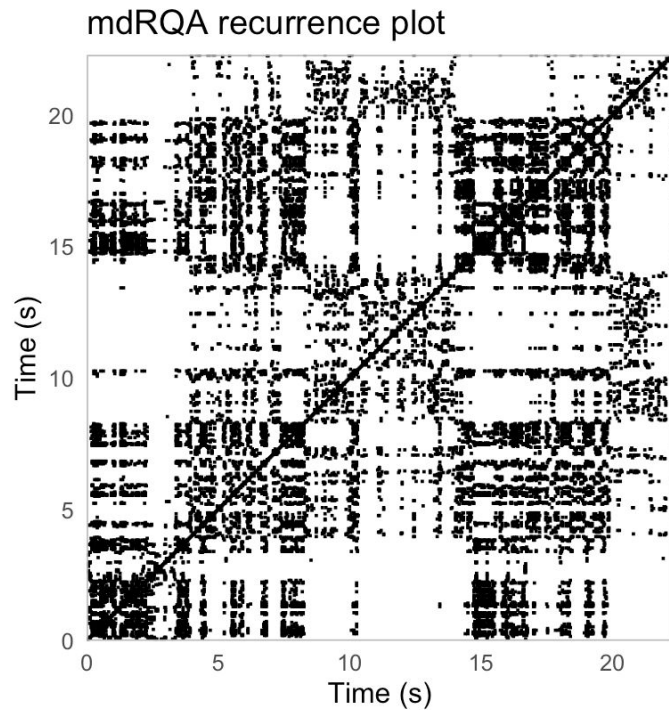
DET LAM
37.10992 51.58283

mdRQA recurrence plot



Affiliated

Results: MdRQA of movement composite



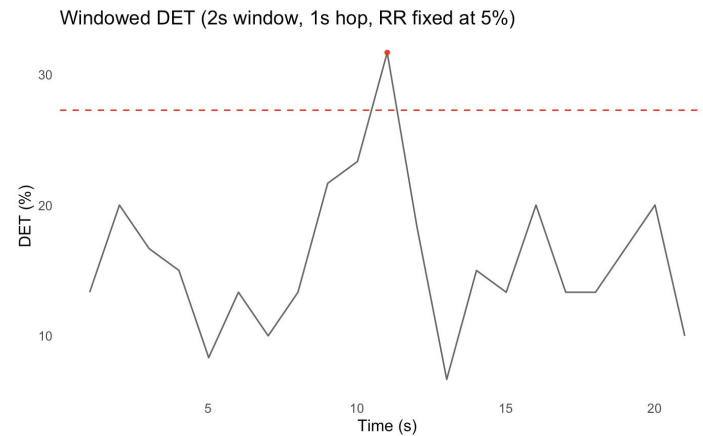
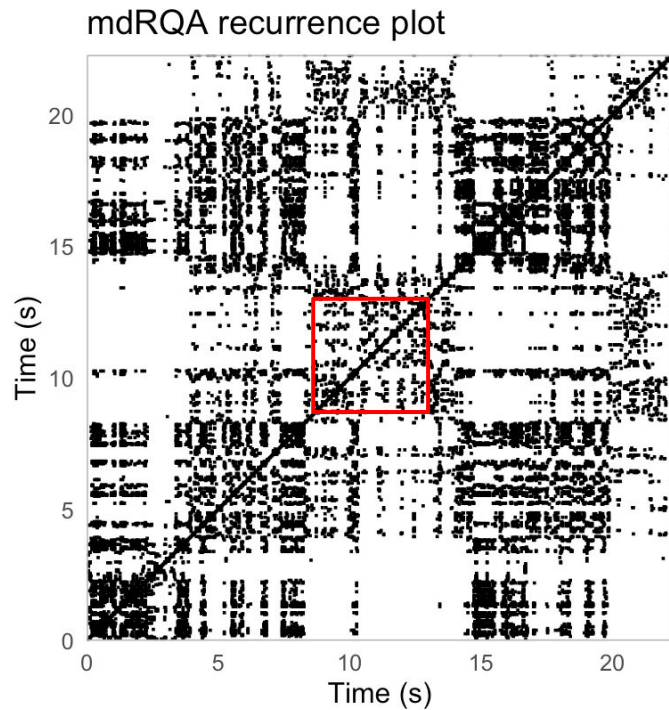
Windowed DET

Slide a window 2 s wide, overlapping by 1 second
calculated %DET from each window

Flagged moments when %DET was +2 sd above
the mean

- i.e., stretches where the dyad's joint movement was more patterned than usual for that trial

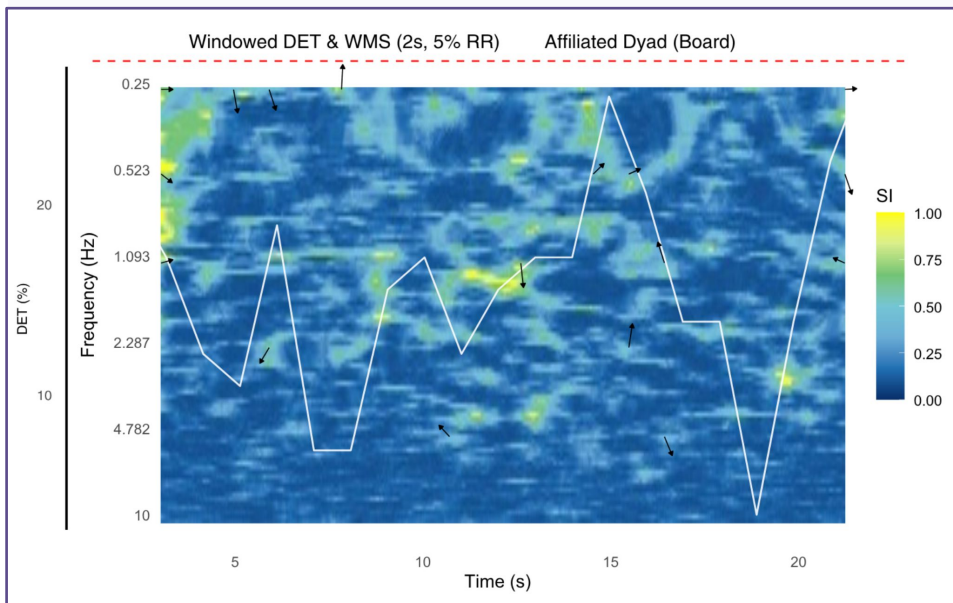
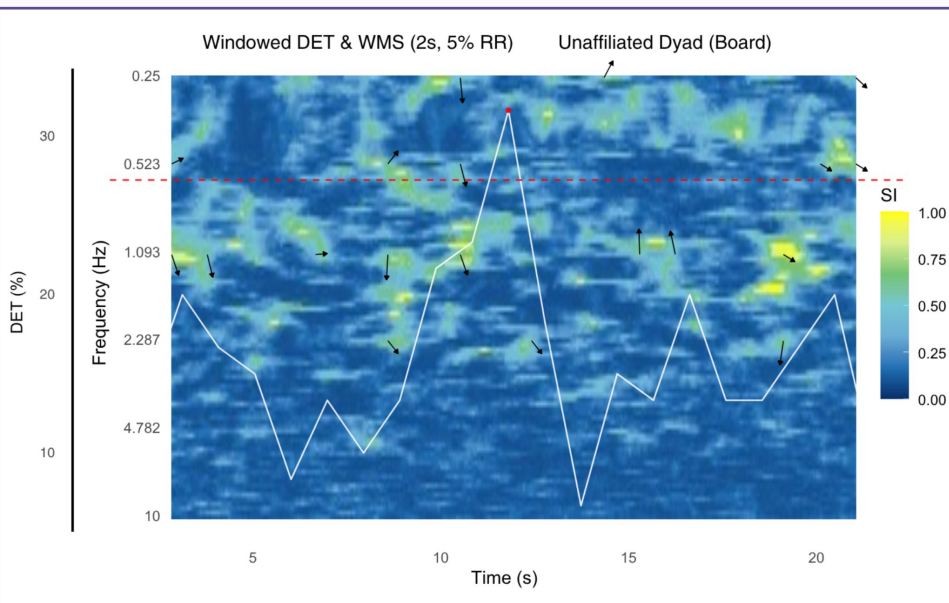
Results: MdRQA of movement composite



Results: Windowed Multiscale Synchrony + Determinism

data_exploration.Rmd

→ in-phase, ← anti-phase, ↑ ↓ lead or lag



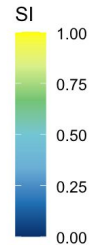
Coordination patterns peak at similar times? Relevant frequencies seem to be in 0.25 - 5 range. No meaningful difference in mean SI! :)

10 Gaze Δx angle timeseries & WMS over 5 trials

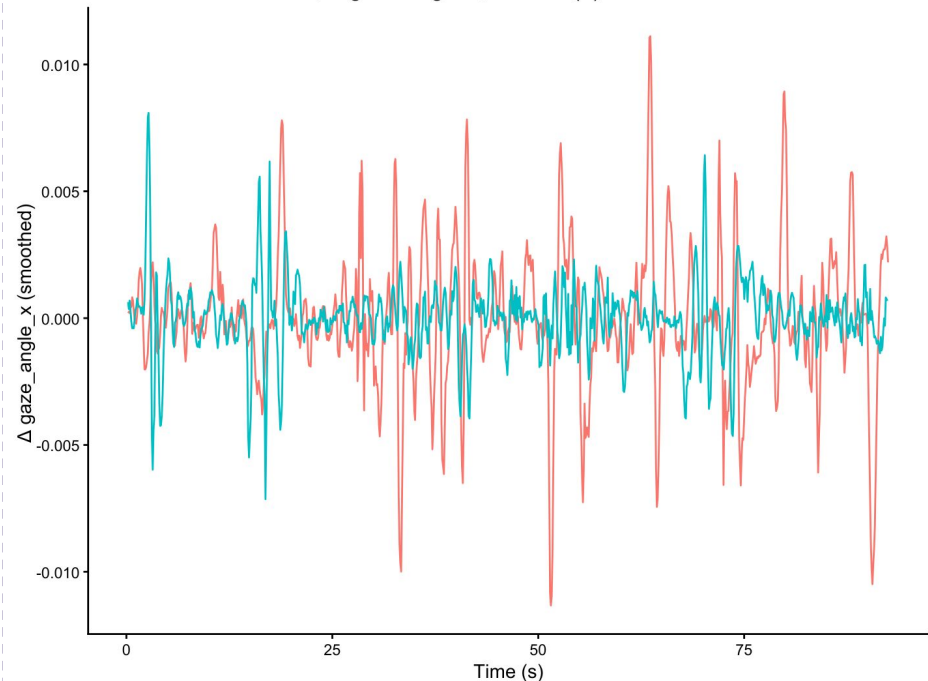
data_exploration.Rmd

→ in-phase, ← anti-phase, ↑ ↓ lead or lag

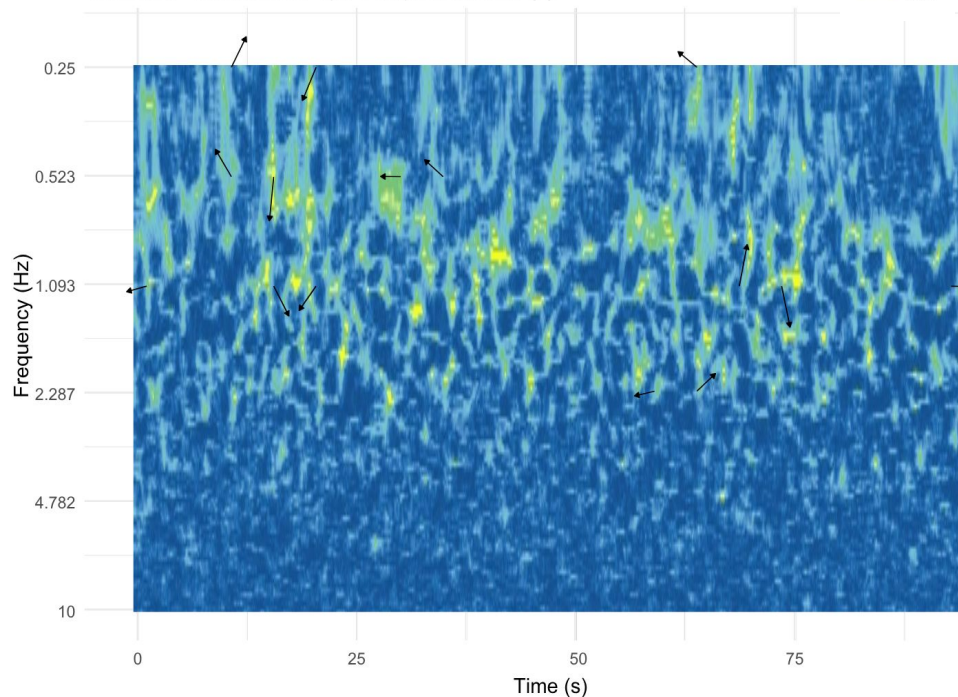
role
— clueGiver
— guesser



On board - unaffiliated, Δ gaze angle x, trials n: (5)



On board - unaffiliated Δ gaze angle x trials n: (5)

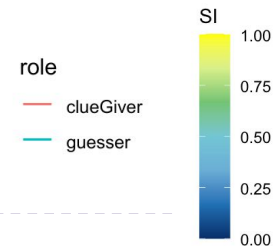


Cyclical patterns that emerge over a longer time window? Fractals?

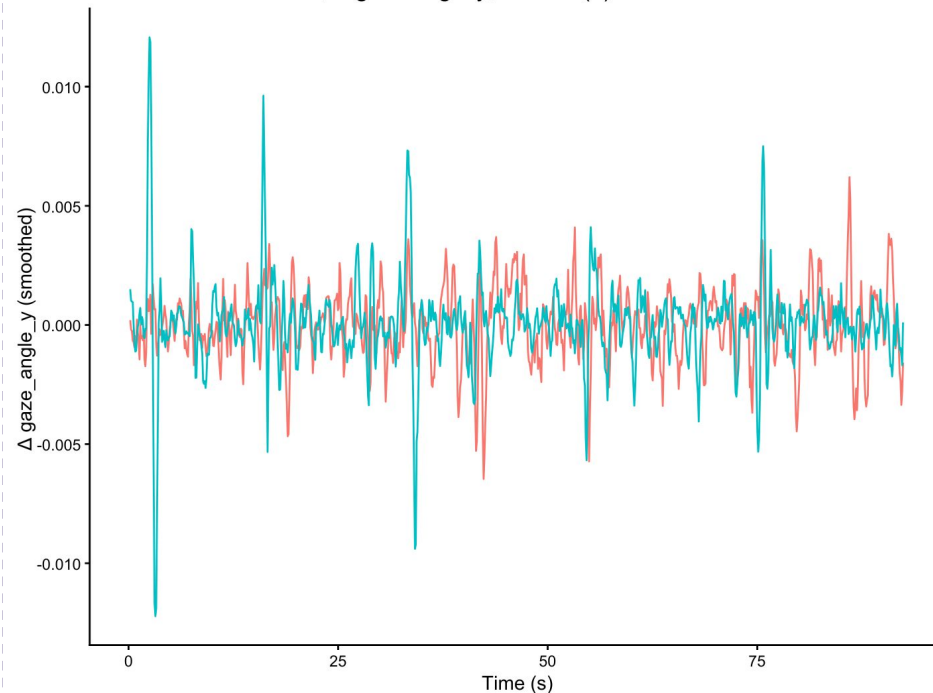
11 Gaze Δy angle timeseries & WMS over 5 trials

data_exploration.Rmd

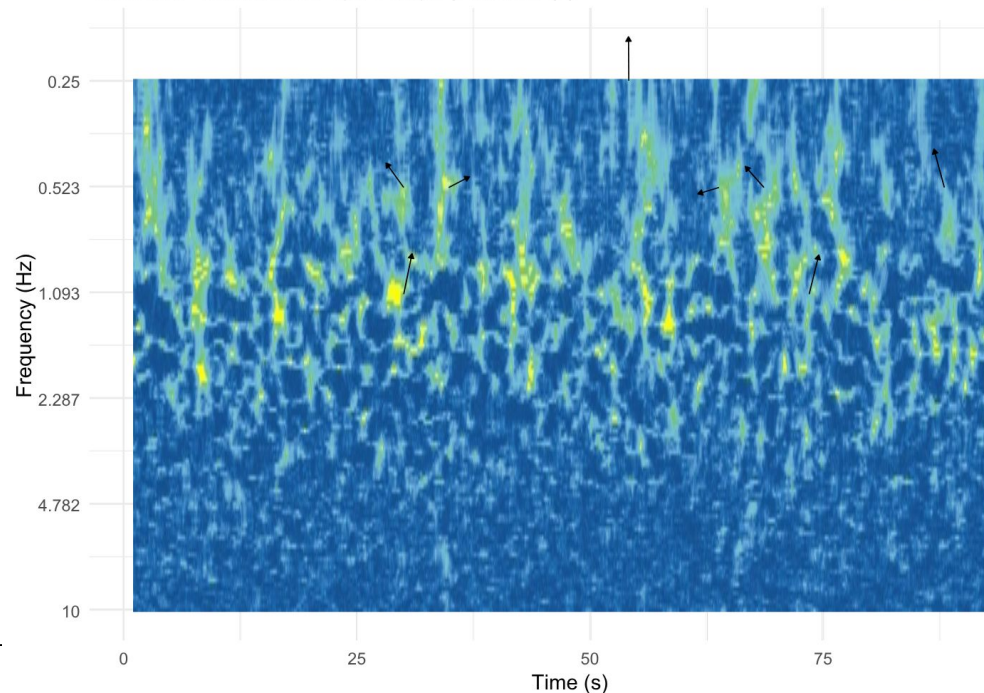
→ in-phase, ← anti-phase, ↑ ↓ lead or lag



On board - unaffiliated, Δ gaze angle y, trials n: (5)



On board - unaffiliated Δ gaze angle y trials n: (5)



Frequency ranges of interest: 0.25 - 3 for gaze → check for more data → narrow the range for mdRQA

From pre-registration to full paper

- 1 **Done — 5 days** Pre-registration · working MDRQA pipeline (movement / gaze) · proof of concept on 2 dyads
- 2 **Refine Coordination Measure** Further exploration is necessary to construct a multimodal measure of coordination within this dataset.

Prior RQ: Do friends coordinate more than strangers?

Does the multi-scale structure of coordination differ between affiliated and unaffiliated dyads?

- 3 **Scale up** Full corpus: 30 dyads = 13 friend / 17 stranger. The teaching subset holds only 2 — affiliation has no variance in it. Precondition for H1, not an option.
- 4 **Confirmatory** Register Hypothesis for MDRQA analysis of full data-set. Mixed-effects: dyad random, affiliation between-dyad, board/ground within. Virtual-pair surrogates throughout — SI and DET are inflated by autocorrelation alone.
- 5 **Cross-scale** WMS computes SI per frequency independently. MFDFA or wavelet bicoherence instead: does one frequency modulate another?

Still missing: an outcome variable — no trial accuracy or solution latency in the metadata, so coordination cannot yet be tied to communicative success. **And a theoretical claim:** coordination is not a property of two people but of a cross-scale, cross-modal coupled system; affiliation modulates its permeability, and perturbation is how we reveal it.

Thank you for your time and attention

Contributions

Role	Who
Data exploration & problem framing	Cathy (Ying) · Ethel · T'kara · Oliver · Zeynep
Literature review	Zeynep
Data analysis	Ethel · T'kara
Slides creation	Cathy (Ying) · Ethel · T'kara · Oliver · Zeynep

Questions?

All code, parameters and decisions are versioned in the group repository

github.com/ethelpruss/purplecircle (not very clean yet)

References

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- Fujiwara, K., Kimura, M., & Daibo, I. (2020). Rhythmic features of movement synchrony for bonding individuals in dyadic interaction. *Journal of Nonverbal Behavior*, 44(1), 173-193.
- Khoramshahi, M., Shukla, A., Raffard, S., Bardy, B. G., & Billard, A. (2016). Role of gaze cues in interpersonal motor coordination: towards higher affiliation in human-robot interaction. *PloS one*, 11(6), e0156874.
- Manrique, H. M., Marín, A., Nieto-Alemán, P. A., Read, D. W., Hernández-Jaramillo, J., García-Palacios, A., & Zeidler, H. (2021). Behavioural mimicry as an indicator of affiliation. *PLoS One*, 16(5), e0250105.
- Marwan, N., Romano, M. C., Thiel, M., & Kurths, J. (2007). *Recurrence plots for the analysis of complex systems. Physics Reports*, 438(5–6), 237–329. <https://doi.org/10.1016/j.physrep.2006.11.001>
- Wiltshire, T. J., Steffensen, S. V., & Fiore, S. M. (2019). Multiscale movement coordination dynamics in collaborative team problem solving. *Applied ergonomics*, 79, 143–151. <https://doi.org/10.1016/j.apergo.2018.07.007>

5 Pipeline — composite movement

1	Input	MediaPipe Pose CSVs. One file per participant per trial.
2	7 (14) core landmarks	Nose, shoulders, elbows, wrists, hip, knees, and ankles
3	Per-frame displacement	Euclidean distance each landmark moves between consecutive frames; only X and Y planes
4	Shoulder-width units	Divided by the clip's median shoulder width — one constant per clip, so the score means the same regardless of distance from camera.

Pipeline — gaze & face

 Import	OpenFace output, cam01 only, "preparation" folders dropped. Folder name parsed into dyad, session_datetime, word, role, cam, condition.
 Movement Deltas	Change from previous frame, within each recording (dyad + session_datetime + role).
 Gaze	gaze_angle x, y — and their frame-to-frame deltas.
 Downsample	Bin into 0.1 s windows on OpenFace's native timestamp → ~10 samples/sec.

Sanity check before every WMS run: trial resolves to exactly one dyad / word / condition, both roles present, and the two cams' start–end timestamps agree.

Where we stand

Subject	Decision	Notes
Dataset	Balance Corpus	—
Camera	Camera 1	Camera 1 is more "straight"; camera 2 sits at an angle
Alignment	Original timestamps	Start/end times & alignment should already be there in the og data

What WMS does not do

- Cross-modal — we explored modalities separately via WMA; it does not account for dependencies across them.
- Cross-scale — WMA does not account for cross-scale dependencies or interactions.
- **SI is computed per modality, per frequency, independently.**

Open

- 1 Does "higher coordination" mean higher mean SI — or peak SI, or SD?
- 2 Composite movement score: which body parts go into it?
- 3 All variables must resolve to the same time series.